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(1) Research Program/Project/Study Title: Earthquake Preparedness of Capiz State University Mambusao Satellite College: An Assessment	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	Mambusao
		Province:	Capiz
		Region:	6
(5) Classification Research: Development: ☒ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Education		(8) Discipline/Field of Specialization Social Sciences/Education	

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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.) This study aimed to evaluate the Earthquake preparedness at Capiz State University (CapSU) Mambusao Satellite College. Utilizing a descriptive research design, the study employed an evaluation form from the university's DRRM, focusing on five phases: Alarm, Building Exit, Evacuation Safe Zone, Room Stability Safety Check, and Technical/Logistics. Data were collected during nationwide simultaneous earthquake drills in the first semester of the 2023-2024 academic year. The results revealed commendable preparedness in the alarm phase, with high scores for siren system availability and student response. However, areas for improvement were identified, particularly in students' execution of safety procedures during alarms. The faculty demonstrated readiness in guiding evacuations, but students' behavior during evacuations indicated a need for enhancement in seriousness and appropriateness. Overall, the Earthquake Drill's Team showed high preparedness in evacuation procedures, room safety checks, and technical/logistics aspects. The study recommends orientation and re-orientation sessions for students, improvements in signage visibility, and consistent maintenance of technical resources. Furthermore, the implementation of targeted plans and programs focusing on emergency protocols and behavioral responses is advised. Future research should explore the interrelationships among DRRM mitigation, response, and rehabilitation levels within the university context. (9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.) Preparedness, DRRM, Assessment	
(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews) In recent times, weather related disasters are the prominent headlines worldwide (Eisensee & Strömberg, 2007), (Walch, 2017). The Philippines being one of the most disaster-prone countries had experienced floods, storms, earthquakes, heat waves and different weather-associated events in the last two decades have induced majority of disasters. Every year there were millions of human beings are afflicted with such disasters there were billions of people injured and displaced because of the effect of disaster. As indicated in World Risk Index Report September 2022, the Philippines is one of the top countries in the world at risk of weather-related disasters. The idea that knowledge of disaster can enhance the preparedness oneself is seen as an important aspect of disaster management. The Functionalist Theory developed by Emile Durkheim (1893) asserts that education meets societal needs by fostering social cohesiveness through developing skills (Prieto, Arcanghel, Corpuz, 2019). Thus, schools' purpose is to prepare students for participation in society's institutions, as well as the transfer of essential values for social responsibility with a goal of socializing people by bringing people from	

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various backgrounds together. Education, according to functionalists, is a valuable contribution to a well-ordered society in which functionalist theory emphasizes social cohesion and solidarity (Prieto, et al.,2019).

As reflected in the CHED Memorandum No. 12 series of 2014 provides the implementing guidelines in accessing the funds to support rehabilitation, and reconstruction works and activities including purchase of equipment and other instructional materials for State Universities and Colleges (SUCs) to implement the appropriate provisions for SUCs of the Rehabilitation and Reconstruction Program (RRP) and the unprogrammed fund of RA 10633 and the supplemental appropriation under RA 10636 to comply with the requirements and processes of the NDRRMC in the rehabilitation and reconstruction per provision of Republic Act 10121 otherwise known as “Disaster Risk Reduction and Management Act of 2010” which mandates all national government agencies to institutionalize policies, structures, coordination, mechanisms and programs with continuing budget appropriation on Disaster Risk Reduction Management (DRRM) from national to local level.

The Capiz State University constitute its Disaster Risk Reduction Management (DRRM) Council as the arm of the university on the preparedness, response, prevention and mitigation, rehabilitation and recovery in times of calamities and disaster (Capiz State University DRRM Plan page 4-5).

As part of the university DRRM Plan, Mambusao Satellite College created a Disaster Risk Reduction and Management (DRRM) Team as the authorize body inside the university who immediately responds to the mass movement, developing an early warning device system and identifying safety evecuation site in order to reduce emergency and disaster risks among its faculty, staff and students. It is on this premise that this study will be conducted to determine the level of preparedness on Earthquake of the Dissaster Risk Reduction and Management (DRRM) Team of CapSU MSC.

Statement of the Problem

This study was conducted to determine the preparedness on Earthquake of the Disaster Risk Reduction and Management (DRRM) Team of Capiz State University Mambusao Satellite College.

Specifically, the study will pursue to answer the following questions:

1. What is the level of preparedness of the Disaster Risk Reduction and Management (DRRM) Team of Capiz State University Mambusao Satellite College?; and
2. What plan of action may be drawn based on the results of study?

(11) Materials and Methods or Methodology (This includes the materials, methods,duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

This study utilized descriptive research design. Descriptive research tries to describe phenomenon, occurrence, event that happens in the present Creswell (1994). In this study, the level of preparedness will be described. This study was conducted at CapSU Mambusao Satellite College, Poblacion, Mambusao, Capiz during the first semester of the school year 2023-2024.

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The research instrument used in the study was an evaluation form from University's DRRM. It consist of five (5) phase. Phase 1 – Alarm Phase , Phase II- Building Exit Phase, Phase III- Evacuation Safe Zone, Phase IV-Room Stability Safety Check and Phase IV- Technicals/Logistics. The results of the analysis of this was based on the data collected from the DRRM panel of evaluators during the conduct of the different drills (earthquake, fire and flood).

The data of this study was gathered based on the evaluation of different evaluators during the Nationwide simultaneous earthquake drill. After retrieving the questionnaires, the data were process and analyzed. In the analysis, descriptive statistical tools used were frequency count, percentage, and mean.

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Level of Preparedness

As shown in table 1, the CapSU MSC is prepared on statements “1. There is an available siren system for the alarm phase”, “2. The siren system is loud enough to be heard”, and “3. The students walk outside upon hearing the siren.” with a mean of 3.0. respectively. This means that the presence of an available siren system for the alarm phase indicates a high level of preparedness. The fact that the siren system is loud enough to be heard signifies another aspect of preparedness, students are appropriately responding to the siren by walking outside. However, on the statement number 4 “The students executed Duck-Cover-Hold during the siren”, result shows that the CapSU MSC is somewhat prepared with a mean of 2.43. This indicate that There's room for improvement in ensuring that students execute the Duck-Cover-Hold procedure effectively upon hearing the siren. The grand mean of 2.86 is verbally interpreted as prepared. This implies that Disaster Risk Reduction and Management (DRRM) Team of CapSU MSC is prepared in terms of the alarm phase.

Table 1 shows the level of preparedness in terms of alarm phase.

Statement	Mean	Interpretation
1. There is an available siren system for the alarm phase.	3.0	Prepared
2. The siren system is loud enough to be heard.	3.0	Prepared
3. The students walk outside upon hearing the siren.	3.0	Prepared
4. The students executed Duck-Cover-Hold during the siren.	2.43	Somewhat Prepared
GRAND MEAN	2.86	Prepared

As shown in table 2, in terms of building exit phase, When it comes to assigning a class team leader to assist the evacuation and guiding students in exiting the building, the faculty are “prepared” with the mean of 3.00. It simply shows that the faculty seem to be more

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prepared in guiding students during the evacuation process. When it comes to students running to exit the building, they are “somewhat prepared” with a mean of 2.57 and students are talking and laughing during the evacuation, are interpreted as “somewhat prepared” with a mean of 1.86. This indicates that students are moderately prepared when it comes to running while exiting the building during an evacuation. Further, students are less prepared in terms of maintaining a serious demeanour during the evacuation. Talking and laughing could indicate a lack of awareness or understanding of the seriousness of the situation.

The grand mean of 2.61 is interpreted as “somewhat prepared” implies that is essential that students are equipped with an in-depth knowledge and understanding in the DRRM competency. There's also a need to improve students' seriousness and preparedness during evacuations. The result of the study was supported by the study of (Mamon et al., 2017) which states that student should understand what makes their school or community unsafe, as well as how they can make these places safe and should possess knowledge on what to do before, during, and after natural disasters that their active student participation and cooperation are clearly critical to the success of DRRM.

Table 2 shows the level of preparedness in terms of building exit phase.

Statement	Mean	Interpretation
1. The students run in exiting the building.	2.57	Somewhat Prepared
2. The faculty guide his/her students in exiting the building.	3.0	Prepared
3. The students are talking and laughing during the evacuation	1.86	Somewhat prepared
4. The faculty assign a class team leader to assist the evacuation.	3.0	Prepared
GRAND MEAN	2.61	Somewhat Prepared

In terms of evacuation safe zone, as shown in table 3, all statements: The class position/ arrangement in order at the evacuation point, the students conducting a headcount at the evacuation point and the faculty checking the students for injury during the shaking has a mean of 3.00 interpreted as “prepared” while Presence of markers or signage at the evacuation point had a mean of 2.86 is also interpreted as “prepared”. This indicates that there are some markers or signage available at the evacuation point. Further, the fact that the class is positioned or arranged in order at the evacuation point signifies a high level of preparedness. Faculty being prepared to check students for injuries during a shaking (presumably during the evacuation) demonstrates a high level of readiness and concern for the well-being of students in a emergency. The grand mean of 2.96 was interpreted as “prepared” indicates that both faculty and students have a commendable level of readiness and proactive measures in place for various aspects of evacuation procedures.

Table 3 shows the level of preparedness in terms of evacuation safe zone

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Statement	Mean	Interpretation
1. There are markers or signage at the evacuation point.	2.86	Prepared
2. The class position/ arrangement in order at the evacuation point.	3.0	Prepared
3. The students conducts a headcount at the evacuation point.	3.0	Prepared
4. The faculty check the students for injury during the shaking.	3.0	Prepared
GRAND MEAN	2.96	Prepared

In terms of room stability safety check, table 4 shows in statements in terms of checking the room for damages conducted by building safety inspection team, reporting the status to the incident commander by the incident management team leaders, coordinating to safety agencies and relying instructions to the faculty by the incident commander, all has a mean of 3.0 interpreted as “prepared”. This simply shows that the team's ability to check the room for damages signifies a high level of preparedness. This indicates that they are equipped and trained to assess structural integrity or potential hazards after an incident, ensuring safety for subsequent actions.

The grand mean of 3.0 indicates that Disaster Risk Reduction and Management (DRRM) Team of CapSU MSC is “prepared.” This indicates a strong emergency response system within the building, characterized by well-prepared teams and leaders.

Table 4 shows the level of preparedness in terms of room stability safety check.

Statement	Mean	Interpretation
1. The building safety inspection team check the room for damages.	3.0	Prepared
2. The Incident Management Team Leaders report the status to the Incident Commander.	3.0	Prepared
3. The Incident Commander coordinate to safety agencies.	3.0	Prepared
4. The Incident Commander relay instruction to the faculty.	3.0	Prepared
GRAND MEAN	3.0	Prepared

In terms of technical/logistics, as shown in table 5, the evaluators perceived the level of preparedness of Disaster Risk Reduction and Management (DRRM) Team of CapSU MSC

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as “prepared” with a mean of 2.89. The presence of VHF Two-way Radio, Handy First-Aid Kit, Megaphone and Evacuation Zone Signage indicates that the DRRM) Team of CapSU MSC is “prepared.” This implies that the evaluators believed that the CapSU MSC had the quality of being prepared in terms of technical and logistics and they are ready to respond in the event or disasters or emergencies.

Table 5 shows the level of preparedness in terms of technicals/logistics.

Statement	Mean	Interpretation
1. Megaphone.	2.86	Prepared
2. VHF Two-way Radio.	3.0	Prepared
3. Handy First-Aid Kit.	3.0	Prepared
4. Evacuation Zone Signage.	2.71	Prepared
GRAND MEAN	2.89	Prepared

Plan of Action

Based on the results of the study, table below shows the implementation plan with its overall expected outcome is to enhance preparedness across different phases, ensuring a well-coordinated and efficient emergency response system at CAPSU MSC.

Phases	Objectives	Strategies/Activities	Persons Involved	Resources	Expected Outcome
Alarm Phase	Enhance execution of Duck-Cover-Hold procedure upon hearing the siren	- Develop and conduct specific training on Duck-Cover-Hold protocols for students and faculty. - Implement regular alarm drills focusing on Duck-Cover-Hold execution. - Conduct evaluations	DRRM Team, CAPSU MSC Faculty, Staff and Students	Training materials, simulation equipment, communication tools	Increased observance and efficiency in executing Duck-Cover-Hold during alarms.

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		post-drills for improvements.			
Building Exit Phase	Improve student behaviour and seriousness during building evacuations.	- Integrate evacuation behavior modules into the curriculum. - Conduct faculty training on maintaining student seriousness during evacuations. - Organize seminars/works hops emphasizing the significance of maintaining decorum during evacuations.	DRRM Team, CAPSU MSC Faculty, Staff and Students	Training modules, seminar materials, engagement platforms	Improved student behavior and seriousness during evacuations.
Evacuation Safe Zone	Enhance the effectiveness of evacuation procedures and resources.	- Assess and upgrade evacuation signage for better visibility and clarity; - Conduct regular training for faculty and students on checkpoint procedures	DRRM Team, CAPSU MSC Faculty, Staff and Students	Signage materials, training materials	Enhanced clarity and effectiveness of evacuation procedures at checkpoints.
Room Stability Safety Check	Maintain high-level readiness for structural safety assessment.	- Regularly update and train inspection teams on safety assessment procedures - Review and update safety assessment protocols based on current best practices.	Safety Inspection Teams, DRRM Team Coordinator	Updated protocols, training materials	Ensured structural safety assessment readiness post-incidents

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Technical/Logistics Readiness	Sustain and improve technical and logistics readiness.	- Implement routine checks and maintenance schedules for technical equipment. - Conduct refresher courses on the usage and handling of technical equipment.	Technical Support Team, DRRM Team	Maintenance schedule, training materials	Consistently maintained and functional technical resources for emergency response.
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(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusion

Based on the results derived from the study, the following conclusions were drawn:

1. CAPSU MSC demonstrates a high level of preparedness in alarm systems and student response but needs improvement in executing safety procedures during alarms. Faculty readiness in guiding evacuations is commendable, but students' behavior during evacuations needs enhancement, emphasizing seriousness and appropriate behavior. The presence of necessary tools and equipment indicates good preparedness.

Recommendations

Based on the results of the study, the following recommendations are made:

1. Results of the study indicated that students are talking and laughing during the evacuation. Thus, the administration is encouraged to conduct orientation and re-orientation of the proper procedures to be taken in exiting the building during unforeseen emergencies and disasters.
2. A provision of improving signage visibility and clarity across evacuation zones and establish consistent maintenance schedules for technical resources is recommended;
3. Implement the target plans and programs for teacher and students emphasizing emergency protocols and behavioural responses.
4. For future researchers, it is recommended to conduct a study on the level of level of preparedness of the Disaster Risk Reduction and Management (DRRM) Team considering its level of mitigation, response, and rehabilitation to determine if there are significant relationships among them.

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(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

- Program implementation plan with its objectives to enhance preparedness across different phases, ensuring a well-coordinated and efficient emergency response system at CAPSU MSC.

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments

This is for the CAPSU MSC Administration, Research Development and Extension Office and the participants. May all our work be appreciated.

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.)

Abs-cbn sagip kapamilya foundation. 2013. Retrieved August 11, 2014 from <http://www.abs-cbnfoundation.com/sk/disaster.html>.

Blakie, P. et al. 1994. At Risk: Natural Hazards, People's Vulnerability and Disasters, Routledge, London, UK.

Cannon, T. 1994. Vulnerability Analysis and Natural Disasters, in A. Varley (ed.), Disasters, Development and Environment, Wiley, West Sussex, UK.

CHED Memorandum No. 12 series of 2014. Guidelines in accessing the funds to support rehabilitation, and reconstruction works and activities including purchase of equipment and other instructional materials for State Universities and Colleges (SUCs).

Delos Santo, rg. & Estolano, EP. 2015. Preparedness and Effectiveness of Disaster Risk Reduction and Management Council of Mambusao, Capiz

Farland M. 2014. Communication towards Effective Disaster Risk Reduction. Retrieved August 12, 2014, from <http://www.ifrc.org/Global/Publications/disasters/reducingrisks/> 302200-Public-awareness-DDR-guide-EN

Human Development Network. 2005. 'Philippine Human Development Report: Peace, Human Security and Human Development', Human Development Network, Quezon City.

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Kuritz, M. 2008. Disaster Preparedness Guide. Retrieved January 12, 2015, from <http://www.effectiveday.com/beingprepared>.

Merriam-Webster Dictionary Nature.com. 2010. Effective Risk Reduction and Response. Retrieved December 18, 2014 from <http://www.nature.com/news/effective-risk-response-needs-a-prepared-mindset-1.16753>.

Philippine Congress. 2010. Disaster Risk Reduction and Management Act of 2010. Retrieved August 5, 2014 from <http://www.dnd.gov.ph/pdf/OCDNDRRMC%20%20Republic%20Act%2010121>.

Thefreedictionary.com

Venton, C. and P. VENTON. 2012. Disaster Risk Reduction and Education. Retrieved from http://learningforpeace.unicef.org/wpcontent/uploads/2012/EEPCT_DRR_CaseStudyA_Philippines_2012.

Wolensky W. 1990. Local Government Roles in Disaster Risk Reduction. Retrieved August 5, 2014, from <http://www.odpm.gov.tt/strategy>

Yodmani, S. 2001. Disaster Risk Management and Vulnerability Reduction: Protecting the Poor. Retrieved from <http://drr.upeace.org/english/documents/References/Topic%205-Risk%20Management>.